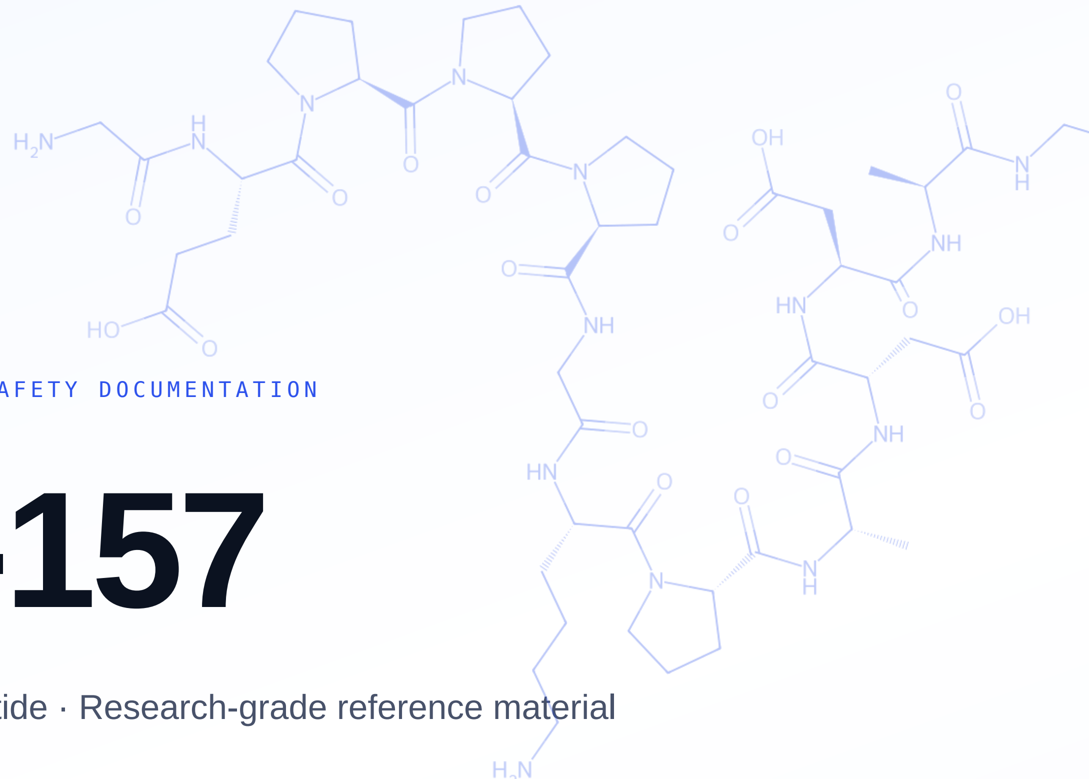
ANVIL COMPOUNDS

RESEARCH USE ONLY

● RESEARCH GRADE · 99%+ PURITY · USA-BASED



RESEARCH REFERENCE & SAFETY DOCUMENTATION

BPC-157

Synthetic pentadecapeptide · Research-grade reference material

CAS	FORMULA	MOL. WEIGHT	CLASS
137525-51-0	$C_{22}H_{98}N_{16}O_{22}$	1419.56 g/mol	Peptide fragment

01 COMPOUND AT A GLANCE

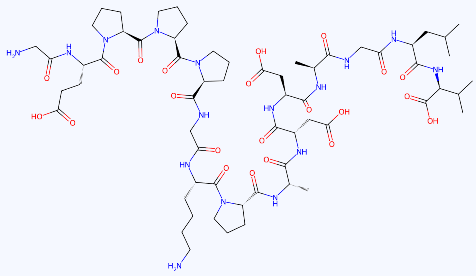
Identity & Composition

BPC-157 is a synthetic 15-amino-acid peptide corresponding to a partial sequence of a protein identified in gastric juice. Supplied as a lyophilized, research-grade reference material.

Product name	BPC-157
Synonyms	Body Protection Compound-157; PL 14736
CAS number	137525-51-0
Molecular formula	$C_{62}H_{98}N_{16}O_{22}$
Molecular weight	1419.56 g/mol
PubChem CID	9941957
Appearance	White lyophilized powder
Solubility	Soluble in water and DMSO
Identified use	In-vitro research & development only

Gly Glu Pro Pro Pro Gly Lys Pro
Ala Asp Asp Ala Gly Leu Val

Primary sequence (N→C): GEPPPGKPADDAGLV



Molecular structure. 2D structure computed from the primary sequence; molecular formula and mass consistent with $C_{62}H_{98}N_{16}O_{22}$ (1419.56 g/mol).

Documentation note. Full hazard, handling, storage and toxicological data appear in the Safety Data Sheet section of this document. The lot-specific Certificate of Analysis (HPLC purity + mass-spectrometry identity) is provided with each order.

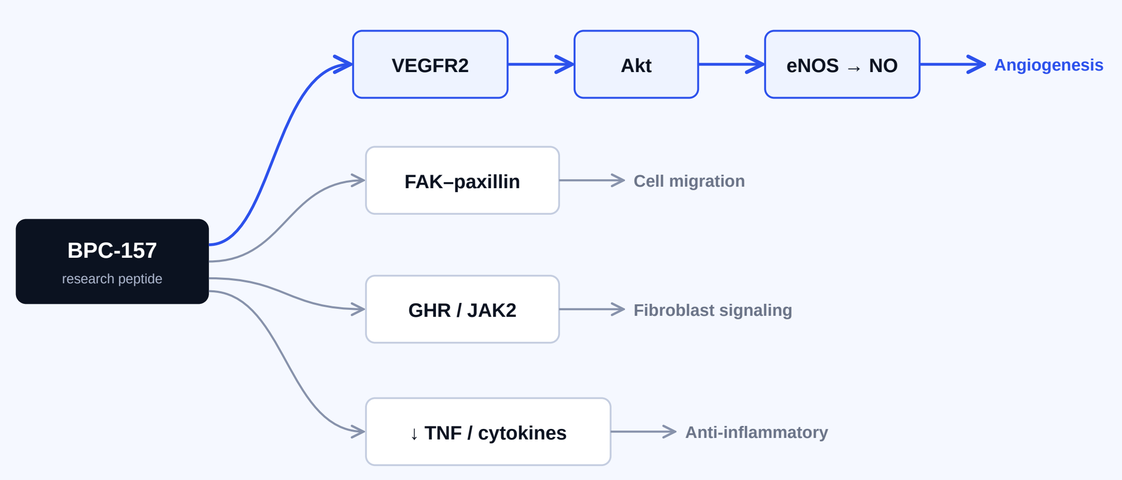
Research Background

Investigated in published preclinical research since the early 1990s across tissue-repair, gastrointestinal, vascular and neurological models. Frequently characterized as notably stable in aqueous and acidic conditions, which is one reason it has been studied across multiple routes of administration in animal models.

Mechanisms studied in preclinical models

Findings below derive from animal and cell-culture systems. They are not established effects in humans.

- **Angiogenesis via VEGFR2 signaling.** Endothelial responses reported through the VEGFR2–Akt–eNOS axis (Hsieh et al.).
- **Nitric oxide (NO) system modulation.** Interaction with the Src–caveolin-1–eNOS pathway; magnitude and direction remain under active debate in the 2025 literature.
- **Growth-hormone receptor (GHR) upregulation.** Increased GHR expression and downstream JAK2 signaling in tendon fibroblast cultures.
- **FAK–paxillin and ERK1/2 signaling.** Implicated in cell-migration and tissue-repair models.
- **Anti-inflammatory modulation.** Reduced pro-inflammatory cytokines in injury models.



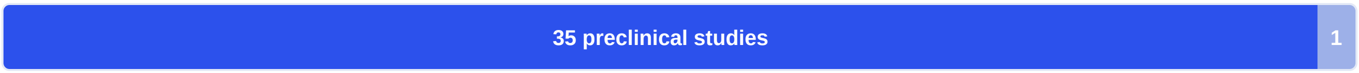
Proposed pathways investigated in preclinical (animal / in-vitro) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn’t

The preclinical mechanistic literature is substantial and, in several pathways, coherent across laboratories. Human data remains very limited. Honest characterization matters for a research reference.

Composition of the reviewed literature

Vasireddi et al. (2025) systematic review, HSS Journal — 544 articles screened (1993–2024), 36 included:



Preclinical (animal / in-vitro): 35 Clinical: 1 (a 2024 IV tolerance pilot — not an efficacy trial)

- The overwhelming majority of data is from **animal and cell-culture models**, not humans.
- Human data is limited to **small pilot studies**, primarily tolerability rather than efficacy, without independent replication.
- **No large, controlled human efficacy trials** have been published.

> **Appropriate characterization:** a peptide with an extensive preclinical research literature and coherent proposed mechanisms, awaiting rigorous human study. Statements that BPC-157 “heals” or “treats” any condition in humans are not supported by current published evidence.

Regulatory & status notes

US FDA	Not approved for any use. Subject to FDA attention in the compounding context (identified as raising safety concerns for compounding use).
WADA	Included on the World Anti-Doping Agency Prohibited List.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Vasireddi N, et al. Emerging Use of BPC-157 in Orthopaedic Sports Medicine: A Systematic Review. HSS Journal. 2025. (PMC12313605)
2. Józwiak M, et al. Multifunctionality and Possible Medical Application of the BPC 157 Peptide — Literature and Patent Review. Pharmaceuticals. 2025;18:185.
3. Hsieh MJ, et al. BPC-157 enhances angiogenesis via the VEGFR2–Akt–eNOS pathway (preclinical). PubMed 27847966.
4. Cushman CJ, et al. Local and Systemic Peptide Therapies for Soft Tissue Injury and Wound Healing. 2024. (PMC11426299)

Safety Data Sheet

Prepared per GHS Rev. 8 / OSHA HazCom 2012
(29 CFR 1910.1200) · 16-section format

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	BPC-157
CAS	137525-51-0
Formula / MW	`C62H98N16O22 / 1419.56`
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	[Supplier entity, address, phone, email]
Emergency	[24/7 monitored contact]

2 Hazard Identification

Not classified based on currently available data. Limited toxicological data; absence of classification is not a determination of absence of hazard.

Signal word: None. **Pictograms:** None required.

Precautionary: P261 avoid breathing dust/aerosols; P264 wash thoroughly after handling; P280 wear gloves/eye protection; P501 dispose per regulations.

3 Composition

Single-substance product. BPC-157 (CAS 137525-51-0), >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water several minutes. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** move to fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. **Physician:** treat symptomatically; no specific antidote known.

5 Fire-Fighting

Media: CO_2, dry chemical, foam, or water spray. Combustion may produce CO, CO_2 and nitrogen oxides. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust formation; ensure ventilation; use PPE per §8. Do not allow entry into drains or waterways. Collect avoiding dust; place in closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive peptide of unknown toxicity; minimize exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; −20 °C long-term, 2–8 °C short-term working stock; keep desiccated. Aliquot reconstituted solutions; store frozen, protected from light. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Local exhaust or fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator per 29 CFR 1910.134 where controls insufficient. Eyewash/safety shower per ANSI Z358.1.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Water; DMSO
MW	1419.56 g/mol
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents, trace transition-metal ions. Decomposition: CO, CO_2, NOx. No hazardous polymerization.

11 Toxicological Information

Not fully characterized. No LD_{50}/LC_{50} available. Skin/eye: treat as potentially irritating. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify requirements.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36) per conditions of use. OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent review by a qualified chemical-safety professional is recommended prior to use. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8; OECD Test Guidelines. Provided without warranty; users responsible for verifying all information and compliance. For scientific R&D use only.

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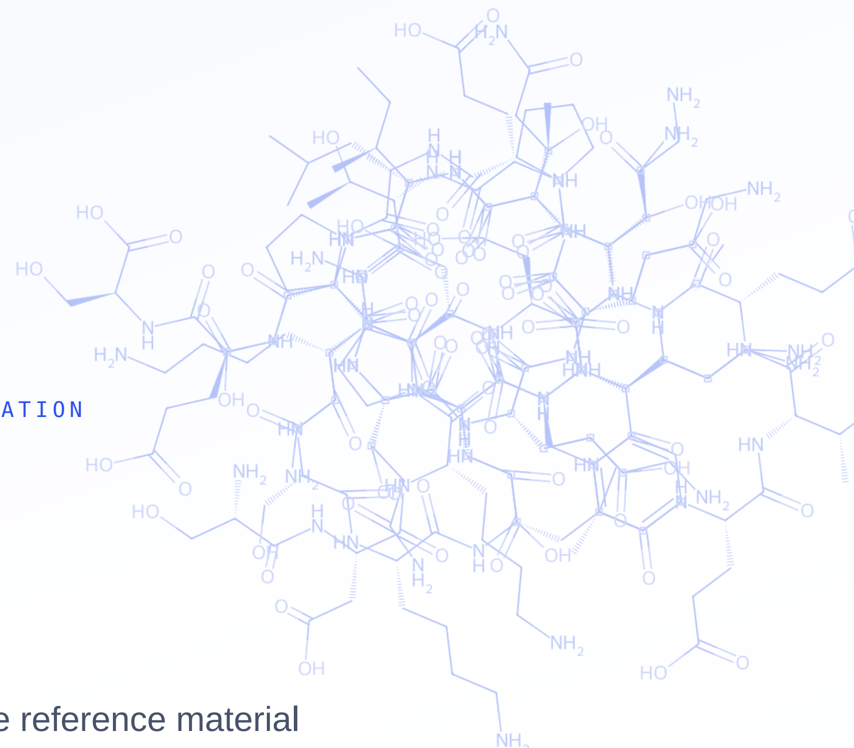


RESEARCH REFERENCE & SAFETY DOCUMENTATION

TB-500

Thymosin β4 fragment · Research-grade reference material

PARENT	ACTIVE DOMAIN	CLASS	FORM
Thymosin β4 (43 aa)	**LKKTETQ**	**Peptide fragment**	**Lyophilized**

Sold for laboratory research purposes only. Not for human or veterinary use, consumption, or therapeutic application. Must be 21+ to purchase. | Independently verified · HPLC + mass spectrometry · Certificate of Analysis per lot.

Identity & Composition

TB-500 is synthetic material based on the actin-binding region of thymosin β4 (Tβ4), a 43-amino-acid actin-regulating peptide found in most human cells. Supplied lyophilized, research grade.

Product name	TB-500
Parent molecule	Thymosin β4 (Tβ4), 43 aa
Parent CAS / MW	77591-33-4 / ≈4921 g/mol
Active domain	LKKTETQ (Tβ4 ~17–23)
Fragment CAS / formula / MW	Depends on product supplied
Appearance	White lyophilized powder
Solubility	Soluble in water
Identified use	In-vitro research & development only

SDKPDMAEIEKFDKSKLKKTETQEKNPLPSKE
TIEQEKQAGES

Thymosin β4 sequence (43 aa) with the actin-binding domain (LKKTETQ) highlighted. TB-500 corresponds to synthetic material based on this active region.

Identity note. In this category "TB-500" is applied inconsistently to both full-length thymosin β4 and a shorter synthetic fragment, and most primary research studies full-length Tβ4. Confirm which your supplier ships and reconcile the CAS / formula / MW accordingly before release.

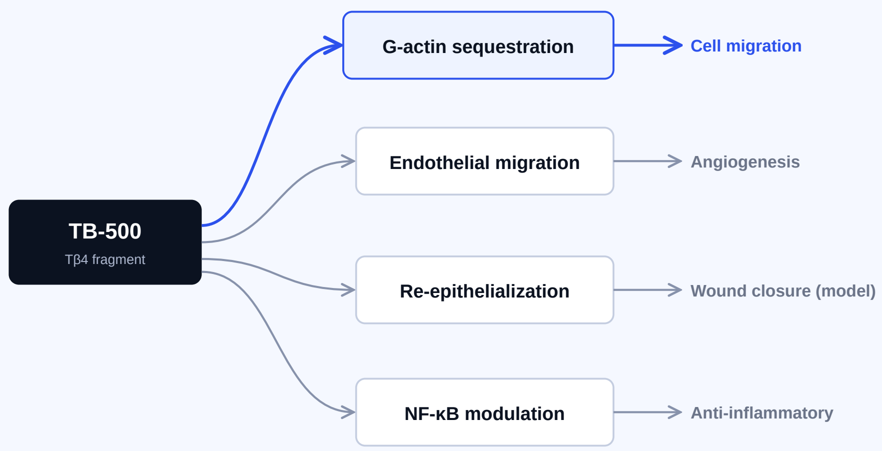
Research Background

Thymosin β4 is the principal intracellular G-actin–sequestering molecule in eukaryotic cells, regulating the actin cytoskeleton that governs cell shape, motility and migration. TB-500 (its active region) has been studied in preclinical tissue-repair research across wound-healing, cardiovascular, corneal and musculoskeletal models since the late 1990s–2000s.

Mechanisms studied in preclinical models

Findings below derive from animal and cell-culture systems. They are not established effects in humans.

- **Actin sequestration (foundational).** High-affinity 1:1 binding to monomeric G-actin (reported Kd ≈ 0.5–0.7 μM), influencing polymerization and cell migration.
- **Cell migration & re-epithelialization.** Promotes keratinocyte and fibroblast migration in preclinical wound models.
- **Angiogenesis.** Endothelial migration and tubule formation; the LKKTET motif is implicated (mutation reduces activity).
- **Anti-inflammatory modulation.** NF-κB-associated modulation of inflammatory signaling in injury models.



Proposed pathways investigated in preclinical (animal / in-vitro) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn’t

The actin-sequestration mechanism and preclinical wound-healing / angiogenesis findings are documented across multiple laboratories. A key nuance: most primary evidence concerns full-length thymosin β4, not the TB-500 fragment as sold by reagent vendors.

Where the evidence sits

Full-length Tβ4 (majority of primary literature) · Direct TB-500 fragment studies (limited; MDPI 2026 scoping review)

- Most robust evidence concerns **full-length Tβ4**; sequence and quality vary between vendor sources of “TB-500.”
- The legitimate clinical path is **RGN-259** (a Tβ4-based ophthalmic candidate) — as of 2026 at phase 3, with **no approved marketed product**.
- **No human randomized controlled trial** supports systemic use for musculoskeletal injury; no regulatory approvals for therapeutic use.

> **Appropriate characterization:** a well-studied actin-regulating peptide with substantial preclinical tissue-repair literature and one active pharmaceutical program (ophthalmology), awaiting approved clinical outcomes.

Regulatory & status notes

US FDA	No approved TB-500 / thymosin β4 therapeutic product. Certain compounded peptides in this class flagged over limited human safety data.
WADA	Thymosin β4 / TB-500 included on the World Anti-Doping Agency Prohibited List.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Thymosin Beta-4 and TB-500 in Tissue Healing, Regeneration, and Musculoskeletal Repair: A Scoping Review. Applied Sciences (MDPI). 2026;16(12):6202.
2. Goldstein AL, Kleinman HK. Thymosin β4 in tissue repair and wound healing (foundational literature).
3. Bock-Marquette I, et al. Thymosin β4 mediates cardioprotective effects (porcine MI model).
4. Malinda KM, et al. Thymosin β4 accelerates wound healing. J Invest Dermatol. 1999.

Safety Data Sheet

Prepared per GHS Rev. 8 / OSHA HazCom 2012
(29 CFR 1910.1200) · 16-section format

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	TB-500 (thymosin β4 fragment)
Parent CAS	77591-33-4 (full-length Tβ4)
Fragment CAS / formula / MW	Depends on product supplied — confirm
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data. Limited toxicological data; absence of classification is not a determination of absence of hazard. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261, P264, P280, P501. Handle as a potentially bioactive peptide of unknown toxicity.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Treat symptomatically.

5 Fire-Fighting

Media: CO_2, dry chemical, foam, water spray. Combustion may produce CO, CO_2, NOx. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive peptide of unknown toxicity; minimize exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; −20 °C long-term, 2–8 °C short-term. Aliquot reconstituted solutions; store frozen. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls insufficient.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Water
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. Decomposition: CO, CO_2, NOx. No hazardous polymerization.

11 Toxicological Information

Not fully characterized. No LD_{50}/LC_{50} available. Skin/eye: treat as potentially irritating. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

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Dispose per applicable regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

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